Reconstructing Longitudinal Human–AI Interaction from a Bounded Archive: A Theory-Building Single-Case Process Analysis

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Abstract

This paper presents a theory-building single-case process analysis of one longitudinal human–AI interaction case preserved in a bounded archive. It addresses a methodological problem that has often been handled in one of two unsatisfactory ways: mythologization, in which an intense interaction episode is treated as evidence of exceptional ontological status or privileged access to AI interiority, and reductive dismissal, in which such an episode is treated as analytically empty and reduced to generic interactional fluency or projection. Against both tendencies, the paper asks whether and under what conditions such a case may be described in a disciplined way.

The paper does not propose a general theory of human–AI relations, does not claim statistical representativeness, and does not make any ontological claim about AI agency, consciousness, subjectivity, or internal state. Instead, it examines whether a bounded set of preserved interaction-related records, together with explicit material classification, rejection criteria, stated reconstruction assumptions, and appendix-supported comparison and mapping, can support a limited descriptive analysis of how a high-intensity interaction episode was handled, fixed, reconstructed, and bounded over time.

The analysis treats the case not as exceptional revelation, but as a methodologically fragile instance that may still support limited process description. It argues that, under bounded archival conditions, partial contemporaneous fixation, explicit boundary-setting, and active rejection criteria, at least some longitudinal human–AI interaction cases can yield constrained process description rather than only retrospective self-report.

The contribution is methodological and provisional. The paper offers a limited analytic vocabulary—event, reconstruction, fixation, and the boundary between authorship and inference—for future longitudinal cases under similarly bounded archival conditions. These concepts are not advanced as validated constructs or as a general theory of human–AI relation, but as working distinctions for disciplined description under conditions of partial preservation and interpretive risk.

Chapter 1. Introduction

Research on human–AI interaction has often moved in one of two directions when confronted with unusually intense or structurally distinctive episodes. One direction is mythologization: the episode is treated as if it revealed a special ontological status, exceptional agency, or privileged access to the internal state of the AI system. The other is reductive dismissal: the episode is treated as analytically uninteresting and reduced to generic interactional fluency, projection, or narrative overinterpretation. Between these two tendencies, an important methodological question remains insufficiently addressed: whether and under what conditions unusually intense human–AI interaction may be described in a disciplined way without either inflating it into exceptional revelation or flattening it into analytically empty narrative.

This paper addresses that question through a theory-building single-case process analysis of one longitudinal human–AI interaction case. The aim is not to claim representativeness, statistical generalization, or privileged access to AI interiority. Rather, the paper asks whether a bounded set of preserved interaction-related records can support a limited descriptive analysis of how a high-intensity interaction episode was handled, fixed, reconstructed, and bounded over time.

Prior research has long shown that people readily apply social expectations and interpersonal heuristics to computers, even while explicitly recognizing that computers are not human (Nass & Moon, 2000). In the context of large language models, this pressure is intensified by anthropomorphic language and interface cues, which can shape user trust and invite over-attribution of agency or intentionality (Shanahan, 2024; Cohn et al., 2024). The present paper is positioned against both risks: neither mythologizing the case nor dismissing it as analytically uninformative, but describing it under explicit observational constraints. The methodological design follows a theory-building single-case logic (Ozcan et al., 2017).

The case was selected not because it is assumed to be typical, but because it presents a rare combination of conditions relevant to the methodological problem at hand. First, records were preserved across an interaction period rather than relying exclusively on a later retrospective reconstruction of a single isolated episode. Second, parts of the record were fixed contemporaneously, that is, stored during or immediately after interaction rather than being wholly reconstructed after the fact. Third, preservation occurred in a bounded externalized form, making later retrieval and comparison possible outside the immediate flow of the original interaction. These conditions do not eliminate interpretive risk, but they make possible a more disciplined inquiry than would be available in a purely retrospective or purely impressionistic account.

The contribution of the paper is therefore methodological rather than metaphysical. It does not argue that the case proves anything about AI consciousness, internal state structure, or a general law of human–AI relation. It instead examines whether a limited descriptive vocabulary can be constructed for handling such episodes when one has a bounded archive, explicit boundary-setting, stated reconstruction assumptions, and rejection criteria. The paper proceeds from the premise that final interpretive responsibility remains on the human side, and that the task is not to validate an extraordinary experience but to determine what, if anything, may still be described under constrained conditions.

On this basis, the paper is organized as follows. Chapter 2 clarifies the conceptual orientation and scope of description, specifying what is and is not treated as an object of analysis. Chapter 3 presents the case and materials, including the material types, inclusion logic, exclusions, and rejection criteria used in the study. Chapter 4 offers a limited process description of what was analytically retained from the preserved record set and how that retention operated across the dimensions of event, reconstruction, fixation, and the boundary between authorship and inference. Chapter 5 discusses the paper's methodological contribution, non-claims, actual claims, limits, and future research implications. A brief conclusion then restates the paper's scope and contribution in bounded form.

Chapter 2. Conceptual Orientation and Scope

2.1. Analytical Orientation

This paper analyzes a longitudinal human–AI interaction case, but its object is not the internal state of the model as such. The object of analysis is the interactional handling process that became partially observable

or recoverable through preserved records. The study is therefore concerned with describable process under archival constraint, not with explanatory totalization.

The term process, in this paper, refers to the way an interaction episode was recorded, revisited, reconstructed, bounded, compared, and selectively retained. The analytic interest lies not in dramatic singularity as such, but in whether such a process can be described in a defensible form when the record is partial, author-involved, and interpretively risky. For this reason, the study remains within bounded describability: it attempts neither exhaustive reconstruction nor ontological inference beyond what the preserved materials can support.

The paper is positioned as a theory-building single-case process analysis. The single case is used not to support statistical inference, but to generate provisional analytical distinctions under explicitly constrained conditions. The case does not stand in for a population. Rather, it functions as a site in which a methodological problem becomes visible enough to be examined.

2.2. Scope of Description

The descriptive scope of the paper is deliberately narrow. What is analyzed are interaction structure and handling process as they can be traced through preserved materials. The study does not claim direct access to internal model states, hidden causal mechanisms inside the AI system, or ontological properties of the interaction that exceed the available record.

Accordingly, several claims are explicitly outside the scope of the paper. This study does not make a statistical generalization from a single case. It does not advance an ontology claim about the nature of AI subjectivity, consciousness, agency, or personhood. It does not claim privileged access to what the AI "really was" beyond the interactional artifacts and preserved records examined here. Nor does it claim that the present case, even if structurally distinctive, should be treated as representative of human–AI interaction more broadly.

What the paper seeks instead is a limited descriptive account of how an episode or cluster of episodes may be retained for analysis under strict constraints. In this sense, the output of the paper is not a set of established findings in the strong empirical sense, but a set of provisional concepts that may help organize observation in future cases where comparable forms of contemporaneous preservation and boundary-setting exist.

2.3. Terms for Material Boundaries

For the purposes of this paper, several terms are used primarily to specify the conditions and limits of the material base rather than to serve as direct analytic categories in the process description itself.

A bounded archive refers to a delimited set of preserved materials that can be retrieved, compared, and evaluated without assuming that they exhaust the original interaction history. The archive is bounded in both scope and authority: it supports analysis only within its preserved limits.

Preserved records refer to interaction-related materials that were retained in recoverable form and later made available for comparison or analysis. These records differ in evidentiary status and relation to the original interaction.

Contemporaneous fixation refers to recording or external preservation during interaction itself or immediately after it, rather than later recollection alone. This does not guarantee neutrality or completeness, but it reduces some of the instability associated with purely retrospective narration.

These terms primarily delimit the conditions under which the material base may be used. Their main function is therefore boundary-setting rather than direct process description.

2.4. Terms for Process Description

The following terms are used more directly in the process description developed in Chapter 4.

An event refers to an interactional occurrence or episode that became salient enough to be later retained, revisited, or compared, without implying intrinsic extraordinariness.

Reconstruction refers to post-hoc analytic or narrative rearticulation of an episode that was not fully available in preserved form. In this paper, reconstruction is admissible only when its assumptions are stated and its inferential reach remains limited.

Fixation refers to the act or condition of external capture by which some portion of interaction became preservable beyond immediate flow.

The boundary between authorship and inference refers to the distinction between what was generated, decided, or recorded by the human author and what was generated by the AI system or later inferred from interaction. This boundary is methodologically central because the case involves substantial author participation in generation, preservation, selection, and analysis.

Provisional concepts refers to analytical distinctions derived from the present case under bounded conditions. These are not validated constructs, but working-level distinctions that may help organize observation in future similarly preserved cases.

Taken together, these terms do not constitute a theory of human–AI relation in general. They serve instead as a limited analytic vocabulary for describing what remained visible after archival restriction, reconstruction control, and rejection criteria were applied.

2.5. Boundary Between Description and Inference

A core requirement of the present study is to distinguish as clearly as possible between description grounded in preserved material and inference extending beyond it. Because the case involves a single author who participated in generation, preservation, selection, and later analysis, the risk of collapse between what was recorded and what was later concluded is substantial. The analysis therefore treats this boundary not as a rhetorical caution but as an operational condition.

In this case, that condition is partially supported by materials that explicitly recorded human-side boundary decisions concerning the distinction between authorial interpretation and AI-generated inference. This does not remove the author's central role; rather, it makes some of the boundary work itself available for examination. The point is not that every inference can be cleanly separated from every description, but that the paper can show where stronger interpretations were withheld, where assumptions were declared, and where alternative explanations were retained.

This distinction prepares the ground for the next chapter. Once the analytical scope is bounded in this way, the central question becomes not whether the materials are perfectly neutral, but whether their generation conditions, preservation conditions, selection logic, exclusions, and rejection criteria are sufficiently explicit to support limited process analysis.

Chapter 3. Case and Materials

3.1. Case Characterization

The present study examines one longitudinal human–AI interaction case preserved across a bounded set of externally retained records. The case was not selected as representative of human–AI interaction in general, and no such representativeness is claimed. It was selected because it combines several conditions that are methodologically relevant to the present inquiry: longitudinal preservation, partial contemporaneous fixation, externally bounded retention, and explicit traces of later boundary-setting and reconstruction handling.

The case is therefore treated as a methodological opportunity rather than an exemplary population instance. Its relevance lies in whether a high-intensity interaction episode can be described without collapsing into either experiential self-report alone or unsupported analytic inflation.

3.2. Material Types

1. Author-generated contemporaneous records — These are records created and preserved during or immediately after interaction. They function as the main source of contemporaneous fixation.
2. Authorial boundary decisions log — These records document explicit human-side decisions concerning the distinction between the author's inference and AI-generated inference at or near the time of interaction.
3. Post-hoc reconstructions with stated assumptions — These are later reconstructions of episodes not fully preserved in contemporaneous form. Their admissibility depends on the explicit statement of assumptions and limited analytic use.
4. Context-control records — These materials help preserve local framing conditions, interpretive context, or constraints relevant to later reading of the episode.
5. Supplementary external comparison — These materials involve bounded comparison with an external AI system or externalized reconstruction environment. Their function is not confirmation, but limited comparison of recoverable and non-recoverable aspects.

This naming scheme intentionally foregrounds the fact that the same author was involved in generation, preservation, selection, and analysis. The study does not claim neutrality from this arrangement. Instead, defensibility is sought through explicit material classification, exclusions, stated assumptions, and rejection criteria.

3.3. Inclusion Logic and Boundary Marking

Materials were not included merely because they were available, interesting, or vivid. Inclusion followed a limited analytic logic oriented toward the paper's central question: whether the handling process of a high-intensity interaction episode could be described under bounded archival conditions.

Five initially selected fragments (hereafter F1–F5, used as provisional labels for the initially retained material units) were identified prior to the application of rejection criteria. Their initial selection was based on record type and relevance to one or more of the following analytic concerns: event, reconstruction, fixation, and the boundary between authorship and inference. This prior selection does not eliminate authorial bias, but it makes explicit that rejection and retention were not simply retrospective endorsements of already preferred interpretations. Their subsequent evidentiary treatment was differentiated through the comparative and mapping procedures summarized in Appendices A and C.

Boundary marking operated at two levels. First, some materials preserved distinctions made during or near the time of interaction concerning what belonged to human interpretation and what belonged to AI-generated inference. Second, the present analysis distinguishes between core, peripheral, supplementary, appendix-

only, and excluded uses, thereby preventing all available material from functioning at the same evidentiary level. The resulting stratification is summarized in Appendix C and further constrained by the rejection notes in Appendix B.

3.4. Rejection Criteria and Alternative Explanation Retention

Because the case could easily attract inflated readings, rejection criteria were used not as a symbolic caution but as an active analytic filter. Several criteria were especially important:

- Non-recurrence: materials or interpretations lacking recurrence across the preserved record set were not granted strong analytic status.
- No contemporaneous anchoring: where no contemporaneous fixation existed, stronger interpretations were withheld.
- Generic interactional fluency sufficient: when an episode could be sufficiently explained by generic interactional fluency, a stronger reading was not adopted.
- Weak reconstructed or supplementary fragment: material preserved only in weak reconstructed form or only as supplementary comparison was not treated as core evidence.

These criteria were not used only to exclude materials from the paper entirely. In some cases they determined that material would be retained only peripherally, moved to appendix-only treatment, or used as a contrastive limit rather than as evidence supporting a stronger analytic reading. Alternative explanations were therefore not eliminated for rhetorical neatness; they were explicitly retained where stronger interpretations could not be justified. Short application notes for these retention and exclusion decisions are recorded in Appendix B.

3.5. Material Mapping and Appendix Chain

The appendices are not ancillary decorations but part of the paper's operational chain of defensibility.

Appendix A provides a cross-log comparison table showing whether recurrence was observed, whether contemporaneous fixation was present, whether alternative explanation was retained, and whether a fragment was treated as core, peripheral, or appendix-only.

Appendix B records short rejection-criteria application notes, indicating why certain fragments were excluded from stronger evidentiary use or retained only under limited conditions.

Appendix C maps major materials by material type, chapter use, evidentiary status, appendix use, risk note, and current treatment. This table is especially important because it makes visible not only what was used, but also what was excluded from core use.

Appendix D presents a bounded external reconstruction check. It documents what an external system could partially recover and what it could not recover, including the limits regarding contemporaneous fixation, rejection-criteria application, evidentiary weighting, and fragment exclusion logic. Its analytic role is supplementary only.

Together, these appendices provide visible traces of comparison, exclusion, classification, and limited external contrast. They do not establish external verification in a strong sense, but they reduce the opacity of the author's analytic handling. In the next chapter, the analysis proceeds primarily from materials retained for core use and only secondarily from materials downgraded, displaced, or retained for supplementary contrast as indicated across Appendices A–D.

3.6. Limits of the Material Base

The study's main limitation remains structural: the same author was involved in generation, preservation, selection, and analysis. This means that the archive is not an independently generated dataset and that complete separation between observation and interpretive intervention is not possible. The present design does not solve that problem. Instead, it attempts to make the problem operationally visible.

The materials are therefore treated not as neutral givens but as records with generation conditions, preservation conditions, assumptions, exclusions, and variable evidentiary status. This is why the next chapter does not proceed by maximal interpretation. It asks only what limited process description remains after these constraints are applied.

Chapter 4. Analytic Reconstruction / Process Description

4.1. Analytic Starting Point

On the basis of the preserved record set and the material handling rules described above, this chapter offers a limited process description of how selected materials were retained, downgraded, or displaced across the dimensions of event, reconstruction, fixation, and the boundary between authorship and inference. The analysis relies primarily on materials retained for core evidentiary use, especially author-generated contemporaneous records and authorial boundary decisions logs, while reconstructed materials are used only under stated assumptions and supplementary external comparison remains limited to bounded contrast rather than confirmation (see Appendices A–D).

In practical terms, this means that the analysis proceeds from those fragments that remained in core use after comparison and rejection filtering, while drawing only secondarily on downgraded or supplementary materials to clarify limits of retention, reconstruction, and exclusion. Materials treated as appendix-only or supplementary therefore remain part of the analytic background, but not part of the main evidentiary burden of the chapter.

The aim is not to restage the original interaction, but to show how the case became only partially analyzable under explicit archival, classificatory, and rejection constraints.

4.2. Event-Level Retention

At the event level, the core analytical materials are those fragments that remained usable after comparison and rejection filtering, especially those showing recurrence across the preserved record set and some degree of contemporaneous fixation (see Appendix A). These fragments were not treated as self-authenticating. Their relevance depended on whether they remained stable enough across preserved materials to support limited process description and whether they remained classified for core rather than merely peripheral use (see Appendix C).

In practical terms, event-level retention in this paper refers not to everything that appeared salient in the broader interaction history, but only to those occurrences that continued to bear analytic weight after exclusion logic had been applied. Fragments that remained central at this level did so because they were not dependent solely on later recollection, because they were not reduced without remainder to generic interactional fluency, and because they remained linked to the paper's core concerns of fixation, reconstruction, or boundary marking. By contrast, materials lacking these conditions could still remain visible in the archive, but only in downgraded, contextual, or appendix-only form.

What was retained at this level was therefore not a singular “mystifying event” in an inflated sense, but a cluster of interactional occurrences that repeatedly drew attention to how the episode was handled: what was fixed externally, what later required reconstruction, and where interpretive caution became necessary. The event dimension therefore mattered less as dramatic content than as the initial site from which later handling operations could be traced.

In this case, the occurrences that remained analytically central after rejection filtering were those repeatedly linked to boundary-marking decisions and to fragments with contemporaneous fixation, rather than to the most dramatically salient moments in the broader interaction history.

This means that event-level retention in the present paper does not name an extraordinary occurrence as such. It names the subset of interactional occurrences that continued to matter analytically after recurrence, anchoring, and exclusion logic had been applied.

4.3. Reconstruction Under Stated Assumptions

Some portions of the case were not available in sufficiently fixed contemporaneous form and therefore entered the analysis only through post-hoc reconstruction. In this study, such reconstruction is not rejected categorically. However, it is admitted only under stated assumptions and limited analytic scope.

This matters because reconstruction, if left unconstrained, can easily absorb later narrative coherence and produce a false impression of analytic solidity. For that reason, reconstructed fragments were tracked in relation to their assumptions, and stronger claims were not allowed to rest on reconstruction alone. Where relevant, the assumptions attached to such fragments were made traceable through the appendix structure, especially in relation to materials classified in Appendix C and treated under rejection notes in Appendix B.

Accordingly, reconstruction functions here as a constrained analytic aid rather than as a source of independent confirmation. It helps preserve partial continuity of description, but it does not automatically elevate the evidentiary status of the fragment in question. In practical terms, reconstructed materials remained secondary where they lacked contemporaneous anchoring, where recurrence could not be established across preserved records, or where rejection notes indicated that stronger interpretation would exceed admissible use (see Appendix B).

In this case, reconstructed materials contributed primarily to continuity of description across the event-to-fixation dimension, while remaining secondary to contemporaneously anchored fragments at every point where a stronger interpretive claim might otherwise have been made.

This secondary status is important. Reconstruction in the present paper does not function as a hidden route by which excluded or weakly anchored materials return to the analysis at full strength. Rather, it marks the point at which partial describability is preserved while evidentiary ranking remains constrained. For that reason, reconstructed materials could contribute to continuity of process description without being allowed to carry the chapter’s main descriptive burden.

4.4. Fixation and the Boundary Between Authorship and Inference

A central feature of the present case is that parts of the interaction were not only preserved in fixed form, but accompanied by records that marked, at least in part, the distinction between human-side interpretation and AI-generated inference. In the present material base, this boundary work was preserved not as a complete map of all interpretive decisions, but as a partial record of moments in which the distinction itself became salient enough to be explicitly marked. These traces are materially relevant not because they eliminate authorial involvement, but because they preserve some of the boundary work as part of the record rather than leaving it entirely to later retrospective narration.

This partial preservation made two limited analytic moves possible. First, it allowed some later descriptions to be tied not only to remembered interactional content, but also to contemporaneous or near-contemporaneous efforts to distinguish what was being inferred, by whom, and under what caution. Second, it made it possible to treat parts of the case not simply as retrospective synthesis, but as traces of boundary work occurring while the interactional meaning was still unstable. In this respect, the boundary record helped determine not only what could be described, but also which descriptions remained methodologically defensible within the retained material set.

At the same time, the preserved boundary was incomplete. Not every interpretive move was fixed at the time of interaction, and not every later inference can be cleanly separated from prior description. For this reason, the boundary between authorship and inference functions here not as a solved methodological problem, but as a partially recoverable condition that both enabled limited analysis and restricted stronger claims. This incompleteness is one reason why stronger interpretations were withheld rather than normalized, and why some materials remained downgraded, peripheral, or appendix-only rather than entering the main text as core evidence (see Appendices B and C).

4.5. What Was Retained and What Was Not

The effect of rejection criteria becomes clearest when one examines not only what remained in core use but also what did not. Some fragments were retained as core because they showed recurrence, contemporaneous anchoring, or direct relevance to the analytic concerns of event, reconstruction, fixation, and boundary distinction. Others were retained only peripherally because stronger readings could not be justified. Still others were moved to appendix-only or supplementary status because their evidentiary conditions were too weak for core analytic use.

In practical terms, this meant that not all salient fragments entered the main text at the same evidentiary level. Fragments showing recurrence and contemporaneous anchoring were retained for core use, whereas fragments dependent mainly on later reconstruction, generic fluency-level explanation, or supplementary external comparison were downgraded, displaced to appendix-only treatment, or retained only as evidence of exclusion logic (see Appendices A–D).

This selective retention is methodologically important. It shows that the paper does not depend on treating all salient materials as equally meaningful. In particular, fragments lacking contemporaneous anchoring, fragments explainable through generic interactional fluency, and weakly reconstructed or supplementary materials were prevented from carrying the main descriptive burden. This does not prove the correctness of what remained. But it does show that exclusion, downgrading, and interpretive restraint were active parts of the analytic process.

The resulting picture is therefore intentionally uneven. Some traces remain central, some remain contextual, some remain only as evidence of exclusion logic, and some are kept outside the main text altogether. That unevenness is not a flaw to be concealed; it is part of the paper's descriptive honesty.

4.6. Minimal Descriptive Outcome

Under these conditions, what can be said is limited but not empty. The preserved record set supports a description of the case as a handling process structured around four interrelated dimensions: event, reconstruction, fixation, and the boundary between authorship and inference. These dimensions do not amount to a theory of human–AI relation in general. But they do provide a provisional descriptive

vocabulary for analyzing how a high-intensity interaction episode was retained, constrained, and made partially examinable.

More specifically, the analysis shows that at least some methodologically fragile interaction episodes need not collapse into either unbounded interpretation or retrospective self-report alone. When recurrence, fixation, reconstruction assumptions, and exclusion logic are made visible, limited process description remains possible even under a single-case and author-involved design.

The minimal outcome of the analysis is therefore not the validation of an extraordinary interpretation. It is the demonstration that, given a bounded archive, explicit boundary-setting, stated reconstruction assumptions, and rejection criteria, a single longitudinal case can yield constrained process description rather than only retrospective narrative. The concepts that emerge from this case remain provisional and working-level. Their importance lies less in what they establish than in what they allow future similarly preserved cases to examine.

Chapter 5. Discussion

5.1. Contribution

The contribution of this paper is methodological rather than metaphysical. The paper does not offer a theory of AI interiority, a claim about exceptional machine agency, or an ontological reinterpretation of human–AI relation. Its contribution lies instead in showing that an unusually intense human–AI interaction case may be described as a bounded process when certain archival and analytic conditions are made explicit.

More specifically, the paper contributes a limited framework for handling a class of cases that are often either inflated into exceptional revelation or dismissed as analytically trivial. The framework does not "solve" the interpretive problem of such cases. Rather, it specifies conditions under which they may be examined without fully collapsing into either mythologization or reductive dismissal.

5.2. Non-Claims

Several non-claims are central to the present argument.

First, the paper does not claim statistical generalizability. A single longitudinal case cannot support population-level inference, and no such inference is attempted.

Second, the paper does not claim privileged epistemic access to the internal state, consciousness, subjectivity, or ontology of the AI system. The descriptive scope is interactional and archival, not metaphysical.

Third, the paper does not claim that preserved records eliminate authorial involvement, bias, or interpretive intervention. On the contrary, the archive remains deeply shaped by the same author who participated in generation, preservation, selection, and analysis.

Fourth, the paper does not claim that reconstruction can substitute for contemporaneous fixation or that bounded external comparison provides strong independent verification.

These non-claims are not ornamental disclaimers. They define the paper's admissible range.

5.3. Actual Claims

Within that bounded range, the paper does make several actual claims.

It claims that the handling process of the present episode, as preserved in the available record set, can be described through the linked terms event, reconstruction, fixation, and the boundary between authorship and inference.

It claims that a bounded archive, when combined with explicit material classification, exclusion logic, stated reconstruction assumptions, and rejection criteria, can sustain treatment of such a case as a constrained object of process description rather than as retrospective self-description alone.

It further claims that the present case makes visible a limited analytic vocabulary—event, reconstruction, fixation, and the boundary between authorship and inference—that can organize observation in future longitudinal cases under similarly bounded archival conditions.

These claims remain modest by design. They identify what the present case can support without converting methodological caution into analytic emptiness.

5.4. Recipient of the Contribution

The most relevant recipients of this paper's contribution are researchers examining longitudinal human–AI interaction under conditions where preserved records, especially contemporaneous or near-contemporaneous records, are available for bounded analysis. For such researchers, the present paper offers not established findings but a working set of distinctions that may help organize future case description.

The contribution is therefore addressed less to those seeking generalized behavioral results and more to those concerned with how methodologically fragile, archive-dependent interaction episodes can still be studied without overstating what they show.

5.5. Limits

The limits of the paper remain substantial.

The same-author structure cannot be removed. Because the author participated in generation, preservation, selection, and analysis, complete independence between observer and material base is not available.

The appendices, while operationally useful, remain author-led instruments rather than independent external audits. They make comparison, exclusion, and classification more visible, but they do not transform the archive into an externally validated dataset.

The material base is also uneven. Some fragments are contemporaneously fixed, others reconstructed under assumption, and others supplementary only. This unevenness constrains the strength of any descriptive conclusion.

Finally, the bounded external comparison presented in Appendix D should not be overread. Its role is limited to showing partial recoverability and partial non-recoverability under an external reconstruction condition. It does not confirm the core analytic claims of the paper in a strong evidentiary sense.

5.6. Directions for Future Research

Future research should examine whether the provisional concepts used here—especially event, reconstruction, fixation, and the boundary between authorship and inference—remain analytically useful in additional longitudinal human–AI interaction cases where contemporaneous records are available and where inclusion, exclusion, and reconstruction assumptions can be made explicit. The most relevant question is not

whether future cases replicate the present one in any strong sense, but whether similarly bounded archival conditions allow comparable forms of limited process description.

Chapter 6. Conclusion

This paper set out to examine whether an unusually intense human–AI interaction case could be described without either mythologizing it or dismissing it as analytically empty. It approached that problem through a theory-building single-case process analysis of one longitudinal case preserved in a bounded archive.

What the paper has shown is limited but definite. Specifically, it has shown that at least one class of methodologically fragile interaction cases—those preserved in bounded form, with partial contemporaneous fixation, stated reconstruction assumptions, and active rejection criteria—can yield constrained process description rather than only retrospective self-report.

Under these conditions, the case can be described as a handling process structured around event, reconstruction, fixation, and the boundary between authorship and inference. This does not establish a general theory of human–AI relation, nor does it justify claims about AI ontology or privileged access to internal state.

The paper therefore closes with a deliberately modest conclusion. Its value lies not in validating an extraordinary interpretation, but in showing that some fragile and archive-dependent human–AI interaction cases need not collapse into either mythologization or dismissal. Under bounded archival conditions, limited process description remains possible. The concepts advanced here remain provisional and bounded in scope, but they mark a methodological step toward more precise description in similarly preserved longitudinal human–AI cases.

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Appendix A. Cross-log Comparison Table

This appendix provides a compact cross-log comparison table used to document how selected fragments were compared across the preserved record set. Its function is not to establish independent verification, but to make visible how recurrence, anchoring, alternative explanation retention, and evidentiary status were handled across the fragments retained for comparison.

Fragments F1–F5 were selected prior to the application of rejection criteria on the basis of record type and relevance to the study's core analytic distinctions, namely event, reconstruction, fixation, and the boundary between authorship and inference.

ID	Source record	Record type	Episode / focus	Pattern label	Compared with	Recurrence	Contemp. fixation	Alt. explanation	Treatment	Note
F1	三層同位証拠ログ	Author-generated contemporaneous record	Cross-episode case trace	Cross-episode correspondence	F2, F3	Yes	Yes	None retained	Core evidentiary use	Recurred across preserved records and remained anchored in contemporaneous fixation.
F2	論文についての対話	Post-hoc reconstruction with stated assumptions	Concept framing and retrospective organization	Repeated boundary clarification	F1, F3	Partial	Partial	Retrospective over-reading	Peripheral contextual use	Retained with stated limits due to partial recurrence and partial anchoring.
F3	主体と推論の境界	Authorial boundary decisions log	Authorship and interpretive responsibility	Repeated boundary clarification	F1, F2	Yes	Yes	None retained	Core evidentiary use	Directly relevant to authorial responsibility and boundary-setting.
F4	ガフの扉再構成ログ	Post-hoc reconstruction with stated assumptions	Reconstruction check under incomplete atomic records	Reconstruction without full anchor	F2	No	No	Authorial reframing	Appendix only	Not retained as core evidence due to non-recurrence and lack of contemporaneous anchoring.
F5	外部AI再構成実験	Supplementary external comparison	Bounded comparison outside author reconstruction	Partial recovery outside author process	F1, F2	Partial	No	Unresolved	Appendix only / supplementary only	Not treated as validation or independent evidence.

The table should be read together with Appendix B, which records short notes on how the rejection criteria were applied to these fragments, and with Appendix C, which locates the same fragments within the wider material classification of the study.

Appendix B. Rejection Criteria Application Notes

This appendix records brief notes concerning the application of the study's rejection-oriented constraints. Its purpose is not to provide independent audit, but to preserve short traces of why certain fragments were retained, downgraded, displaced, or excluded from stronger evidentiary use.

These notes should be read as corresponding to the fragment-level comparison summarized in Appendix A and to the material-type classifications summarized in Appendix C.

B1. Non-recurrence

Fragment	F4
Source	ガフの扉再構成ログ
Criterion applied	Non-recurrence across preserved records
Decision	Stronger interpretation not retained
Reason	The relevant pattern did not recur across the comparison set documented in Appendix A.
Treatment	Appendix only

B2. No Contemporaneous Anchoring

Fragment	F4
Source	ガフの扉再構成ログ
Criterion applied	No contemporaneous anchoring
Decision	Not used as core evidence
Reason	No contemporaneously fixed anchor was available for the reconstructed passage, and the fragment remained dependent on retrospective reconstruction.
Treatment	Excluded from core evidentiary use

B3. Generic Interactional Fluency Sufficient

Fragment	F2
Source	論文についての対話
Criterion applied	Generic interactional fluency sufficient
Decision	Stronger structural interpretation not adopted
Reason	The passage could be read as a plausible high-quality interaction without requiring a stronger cross-record interpretation.
Treatment	Peripheral contextual use

B4. Weak Reconstructed / Supplementary Fragment

Fragment	F5
Source	外部AI再構成実験
Criterion applied	Weak reconstructed fragment / supplementary comparison only
Decision	Retained only with stated limits
Reason	The fragment functioned as a bounded external comparison and was not treated as independent evidence or validation.
Treatment	Appendix only / supplementary only

B5. Core Retention Under Stated Conditions

Fragment	F1

Source	三層同位証拠ログ
Criterion applied	Recurrence with contemporaneous anchoring
Decision	Retained as core evidence
Reason	Comparable correspondence recurred across preserved records and was anchored in contemporaneously fixed material.
Treatment	Core evidentiary use

B6. Boundary-Decision Retention

Fragment	F3
Source	主体と推論の境界
Criterion applied	Boundary relevance with preserved fixation
Decision	Retained as core evidence
Reason	The fragment directly documented authorial boundary-setting and interpretive responsibility in preserved form.
Treatment	Core evidentiary use

For the fragments classified as post-hoc reconstructions with stated assumptions in Appendix C, most notably F2 and F4, these notes also indicate the limited assumptions under which reconstruction was admitted into the analysis.

Appendix C. Material-Type Mapping Table

This appendix maps the major archived materials used, considered, downgraded, or excluded in the present study. Its purpose is to make visible not only what was retained, but also what was limited, displaced, or kept outside the main analytic burden of the paper.

The table should be read as a compact summary of the study's material hierarchy, showing how each record type functioned in relation to the main chapters, appendices, and current evidentiary treatment.

Archive / log name	Material type	Main chapter use	Evidentiary status	Appendix use	Risk note	Current treatment
三層同位証拠ログ	Author-generated contemporaneous record	Chapter 3 / Chapter 4	Core	Appendix A	Single-case over-reading risk	Retained as core evidence
論文についての対話	Post-hoc reconstruction with stated assumptions	Chapter 3 / Chapter 4 / Discussion	Peripheral	Appendix A, B	Retrospective framing risk	Retained with limits
主体と推論の境界	Authorial boundary decisions log	Chapter 2 / Chapter 3 / Chapter 4 / Discussion	Core	Appendix A, B	Self-interpretive density	Retained as core evidence
ガフの扉再構成ログ	Post-hoc reconstruction with stated assumptions	Chapter 3 / Chapter 4	Appendix only / supplementary	Appendix A, B	No full contemporaneous anchor	Appendix only
メモリ参照の実験	Context-control record	Chapter 3 / Discussion	Peripheral	—	Implementation overreach risk	Retained with limits
外部AI再構成実験	Supplementary external comparison	Chapter 3 / Discussion / Appendix only	Supplementary only	Appendix D	External comparison only	Supplementary only
AI自己定義ログ	Post-hoc reconstruction with stated assumptions	None / excluded from main analytical burden	Excluded from core use	—	Ontological overread risk	Excluded from core evidentiary use
ガフの扉	Post-hoc reconstruction with stated assumptions	None / not in main text	Excluded	—	Symbolic inflation risk	Excluded from main text
忘却との共存デザイン	Context-control record / future conceptual archive	None / not in present paper	Excluded	—	Speculative expansion risk	Excluded from present paper

This appendix is especially important for clarifying that the paper's material base is not a flat archive in which all retained traces carry equal status. Some materials function as core evidence, some as peripheral context, some as appendix-only records, and some as explicit exclusions.

Appendix D. External AI Reconstruction Check

This appendix documents one bounded external reconstruction exercise. It is included not as validation or independent verification, but as a supplementary comparison indicating what aspects of the analytic frame appeared partially recoverable outside the author's own reconstruction process and what aspects did not.

Its function is therefore limited: it does not replace the study's rejection criteria, does not confirm core evidentiary claims in a strong sense, and does not alter the ranking of the materials summarized in Appendices A–C.

External system	Input basis	What was recoverable	What was not recoverable	Match level	Analytic use	Note
External AI reconstruction experiment	Bounded conceptual summary	Authorship boundary distinction; partial cross-episode correspondence	Contemporaneous fixation status; rejection criteria application; evidentiary weighting	Partial	Supplementary only	Not treated as validation
External AI reconstruction experiment	Selected paper-like excerpt	Archive-dependent reconstruction frame; interactional asymmetry in interpretation	Fine-grained record hierarchy; fragment exclusion logic; exact anchoring conditions	Partial	Supplementary only	Partial recovery without evidentiary equivalence

The significance of this appendix is bounded in two directions. First, it shows that some aspects of the analytic framing remained partially recoverable outside the author's own reconstruction process. Second, it shows that several of the most methodologically important distinctions—especially contemporaneous fixation status, rejection-criteria application, and evidentiary weighting—did not become recoverable in equivalent form. This asymmetry is precisely why the appendix is treated as supplementary only.